

APPLIED REGRESSION ANALYSIS

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DESCRIPTION:

The dataset contains characteristics of diamonds, including their physical attributes (such as carat, cut, colour, clarity, dimensions) and their price. Diamond pricing is influenced by multiple factors, and multiple linear regression (MLR) can help quantify the relationship between these attributes and the price.

This analysis will help us understand which factors most significantly contribute to the price of a diamond and by how much.

OBJECTIVE:

The main objective of this study is to:

1. **Identify the key factors** (independent variables) that influence the price of a diamond.
2. **Quantify the impact** of each factor on the price using multiple linear regression.
3. **Develop a predictive model** that can estimate the price of a diamond based on its attributes.

VARIABLES:

Dependent Variable (Y):

- **price** → The price of the diamond in US dollars.

Independent Variables (X):

- **carat** → Weight of the diamond (most important factor in pricing).
- **cut** → Quality of the cut (Fair, Good, Very Good, Premium, Ideal).
- **colour** → Diamond colour (graded from J (worst) to D (best)).
- **clarity** → Measurement of diamond purity (I1, SI2, SI1, VS2, VS1, VVS2, VVS1, IF).
- **depth** → Total depth percentage ($z / \text{mean}(x, y) * 100$).
- **table** → Width of top of diamond relative to widest point.
- **x** → Length in mm.
- **y** → Width in mm.
- **z** → Depth in mm.

IMPORTING DATASET:

```
> data = read.table(file.choose() , header = TRUE , sep = ",")
> head(data)
```

	X	carat	cut	color	clarity	depth	table	price	x	y	z
1	1	0.23	Ideal	E	SI2	61.5	55	326	3.95	3.98	2.43
2	2	0.21	Premium	E	SI1	59.8	61	326	3.89	3.84	2.31
3	3	0.23	Good	E	VS1	56.9	65	327	4.05	4.07	2.31
4	4	0.29	Premium	I	VS2	62.4	58	334	4.20	4.23	2.63
5	5	0.31	Good	J	SI2	63.3	58	335	4.34	4.35	2.75
6	6	0.24	Very Good	J	VVS2	62.8	57	336	3.94	3.96	2.48

CLEANING DATASET:

```
> #CLEANING THE DATA
> colSums(is.na(data1))
      X   carat   cut   color clarity   depth   table   price      x      y      z
0      0      0      0      0      0      0      0      0      0      0
width length height
0      0      0

>
> clean_data = na.omit(data1)
> head(clean_data)
  X carat      cut color clarity depth table price      x      y      z width length height
1 1 0.23    Ideal     E    SI2  61.5   55  326 3.95 3.98 2.43 3.95 3.98 2.43
2 2 0.21  Premium     E    SI1  59.8   61  326 3.89 3.84 2.31 3.89 3.84 2.31
3 3 0.23     Good     E    VS1  56.9   65  327 4.05 4.07 2.31 4.05 4.07 2.31
4 4 0.29  Premium     I    VS2  62.4   58  334 4.20 4.23 2.63 4.20 4.23 2.63
5 5 0.31     Good     J    SI2  63.3   58  335 4.34 4.35 2.75 4.34 4.35 2.75
6 6 0.24 Very Good     J   VVS2  62.8   57  336 3.94 3.96 2.48 3.94 3.96 2.48
```

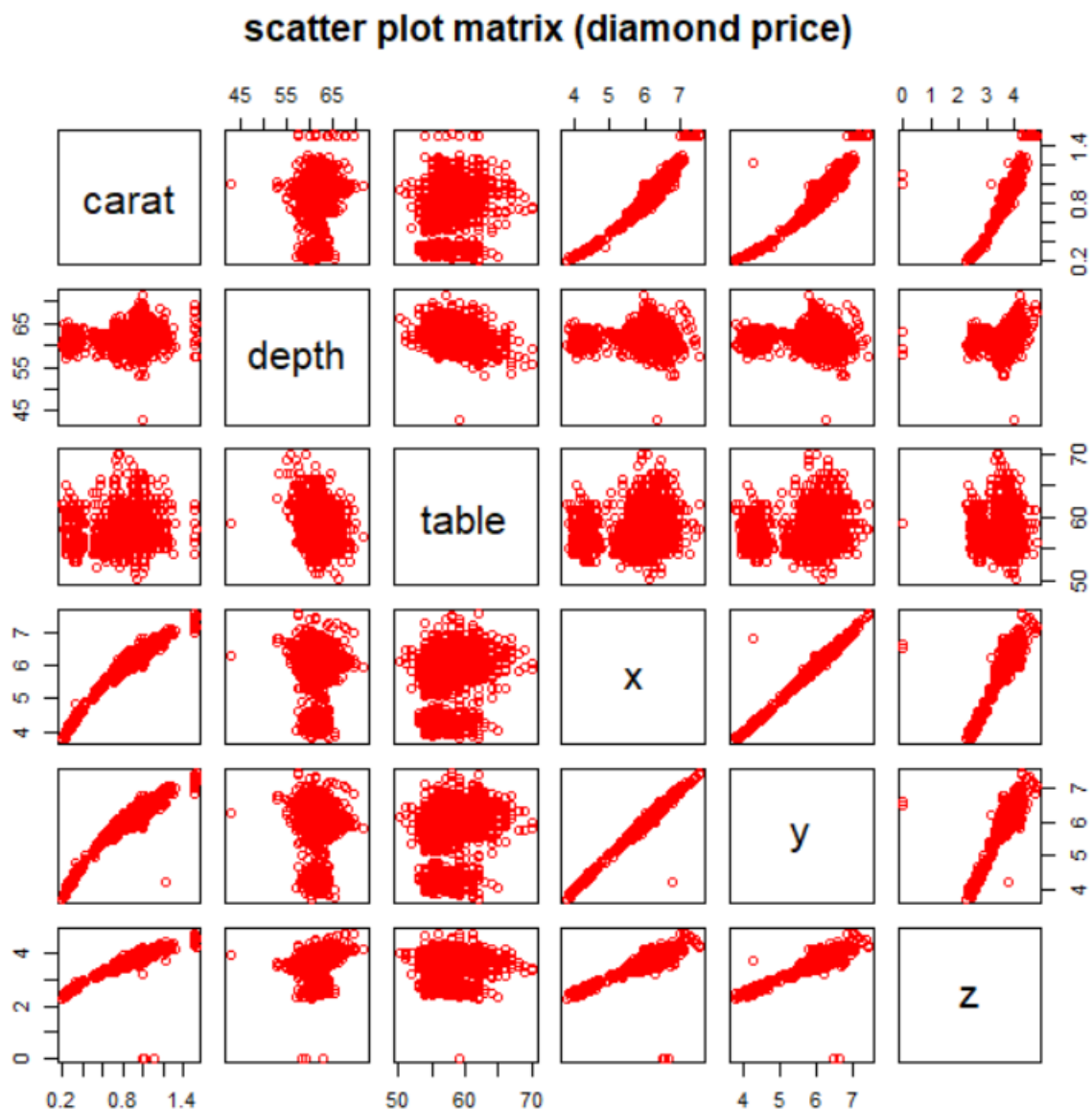
SUMMARY:

```
> summary(data)
      X      carat      cut      color
Min.   : 1    Min.   :0.2000 Length:53940 Length:53940
1st Qu.:13486 1st Qu.:0.4000 Class :character Class :character
Median :26971 Median :0.7000 Mode  :character Mode  :character
Mean   :26971 Mean   :0.7979
3rd Qu.:40455 3rd Qu.:1.0400
Max.   :53940 Max.   :5.0100

      clarity      depth      table      price      x
Length:53940 Min.   :43.00 Min.   :43.00 Min.   : 326 Min.   : 0.000
Class :character 1st Qu.:61.00 1st Qu.:56.00 1st Qu.: 950 1st Qu.: 4.710
Mode  :character Median :61.80 Median :57.00 Median : 2401 Median : 5.700
Mean   :61.75 Mean   :57.46 Mean   : 3933 Mean   : 5.731
3rd Qu.:62.50 3rd Qu.:59.00 3rd Qu.: 5324 3rd Qu.: 6.540
Max.   :79.00 Max.   :95.00 Max.   :18823 Max.   :10.740

      y      z
Min.   : 0.000 Min.   : 0.000
1st Qu.: 4.720 1st Qu.: 2.910
Median : 5.710 Median : 3.530
Mean   : 5.735 Mean   : 3.539
3rd Qu.: 6.540 3rd Qu.: 4.040
Max.   :58.900 Max.   :31.800
```

SCATTER MATRIX PLOT:



Key Interpretations

1. Carat vs Dimensions (x, y, z):

- There is a **strong positive linear relationship** between **carat** and the diamond's physical dimensions (**x, y, z**).
- As the weight (carat) increases, the size of the diamond (length, width, depth) also increases.
- This suggests carat is highly correlated with x, y, and z.

2. x, y, z (dimensions among themselves):

- Strong linear correlation among **x, y, z**.
- These variables are almost linearly dependent, meaning multicollinearity is likely if all are included in regression.

- Example: larger x (length) is always associated with larger y (width) and z (depth).

3. **Carat vs Depth & Table:**

- **Carat vs depth:** No strong trend, more scattered. Depth does not vary much with carat.
- **Carat vs table:** Weak relationship, wide scatter. Table width is not strongly related to carat.

4. **Depth vs Table:**

- No clear linear relationship; both appear widely spread with no pattern.
- Suggests **depth and table are independent variables** that don't strongly correlate with the diamond's weight/size.

5. **Clusters:**

- Some scatterplots show clusters or vertical "bands" (e.g., in **depth** and **table**).
- This indicates many diamonds have standard depth/table percentages (cut standards in the diamond industry).

BUILDING REGRESSION MODEL:

HYPOTHESIS:

Null Hypothesis (H_0): None of the variables (carat, depth, table, x, y, z) have any effect on diamond price.

Alternative Hypothesis (H_1): At least one of the variables (carat, depth, table, x, y, z) does have an effect on diamond price.

```
> # Multiple regression
> mlr = lm(price ~ carat + depth + table + x + y + z , data = clean_data)
> summary(mlr)

Call:
lm(formula = price ~ carat + depth + table + x + y + z, data = clean_data)

Residuals:
    Min       1Q   Median       3Q      Max
-1661.2  -225.8   -15.7   218.3  4358.8

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -21249.343    473.426  -44.884 < 2e-16 ***
carat       -6509.910    141.853  -45.892 < 2e-16 ***
depth         129.517     4.859   26.656 < 2e-16 ***
table         12.465     2.249    5.541 3.15e-08 ***
x           1568.341     90.126   17.402 < 2e-16 ***
y           1838.146     77.031   23.862 < 2e-16 ***
z            186.876     47.586    3.927 8.71e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 333.6 on 4993 degrees of freedom
Multiple R-squared:  0.8557,    Adjusted R-squared:  0.8556
F-statistic: 4936 on 6 and 4993 DF,  p-value: < 2.2e-16
```

Model Summary

- **Dependent Variable:** Price of the diamond
- **Independent Variables:** Carat, Depth, Table, x, y, z
- **$R^2 = 0.8557$** → The model explains **85.57% of the variation** in diamond prices.
- **Adjusted $R^2 = 0.8556$** → Very close to R^2 , meaning the model is not overfitting with extra predictors.
- **F-statistic (4936 on 6, 4993 DF, $p < 2.2e-16$)** → The overall regression model is highly significant.

Important Notes

- The **negative coefficient for carat** is counterintuitive. Normally, carat should positively influence price.
- This happens because **carat, x, y, and z are highly correlated** (multicollinearity). The model struggles to separate their individual effects.

- In practice, we might:
 - Use only **carat** as a main predictor (and drop x, y, z).
 - Or apply **Principal Component Analysis (PCA)** to combine x, y, z into a single size dimension.

For the **overall regression model**, the test checks whether all predictors together explain variation in diamond price. Since the p-value is near zero, we reject the null hypothesis → the model is statistically significant overall.

For the **individual predictors**, each hypothesis test checks if a single variable (carat, depth, table, x, y, or z) influences price when the others are held constant. Because all p-values are very small, we reject all null hypotheses → every predictor has a significant effect on diamond price.

TRANSFORMATION: Since diamond price is highly skewed, I applied a **logarithmic transformation** to the dependent variable (price). This transformation reduces skewness, stabilizes variance, and improves the fit of the regression model. In the log-transformed model, the coefficients are interpreted in terms of **percentage changes in price** rather than absolute dollar changes, making the results more stable and meaningful.

AFTER LOGARTHAMIC TRANSFORMATION.....

```
> log_mlr = lm(log(price) ~ carat + depth + table + x + y + z , data = clean_data)
> summary(log_mlr)

Call:
lm(formula = log(price) ~ carat + depth + table + x + y + z,
    data = clean_data)

Residuals:
    Min       1Q   Median       3Q      Max
-1.24071 -0.09668 -0.00426  0.09779  2.71464

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -10.727376   0.254463  -42.157 < 2e-16 ***
carat       -5.479293   0.076245  -71.864 < 2e-16 ***
depth        0.103890   0.002612   39.780 < 2e-16 ***
table        0.012198   0.001209   10.089 < 2e-16 ***
x            1.542844   0.048442   31.849 < 2e-16 ***
y            1.081275   0.041404   26.115 < 2e-16 ***
z            0.125231   0.025577    4.896 1.01e-06 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1793 on 4993 degrees of freedom
Multiple R-squared:  0.9005,    Adjusted R-squared:  0.9004
F-statistic: 7535 on 6 and 4993 DF,  p-value: < 2.2e-16
```

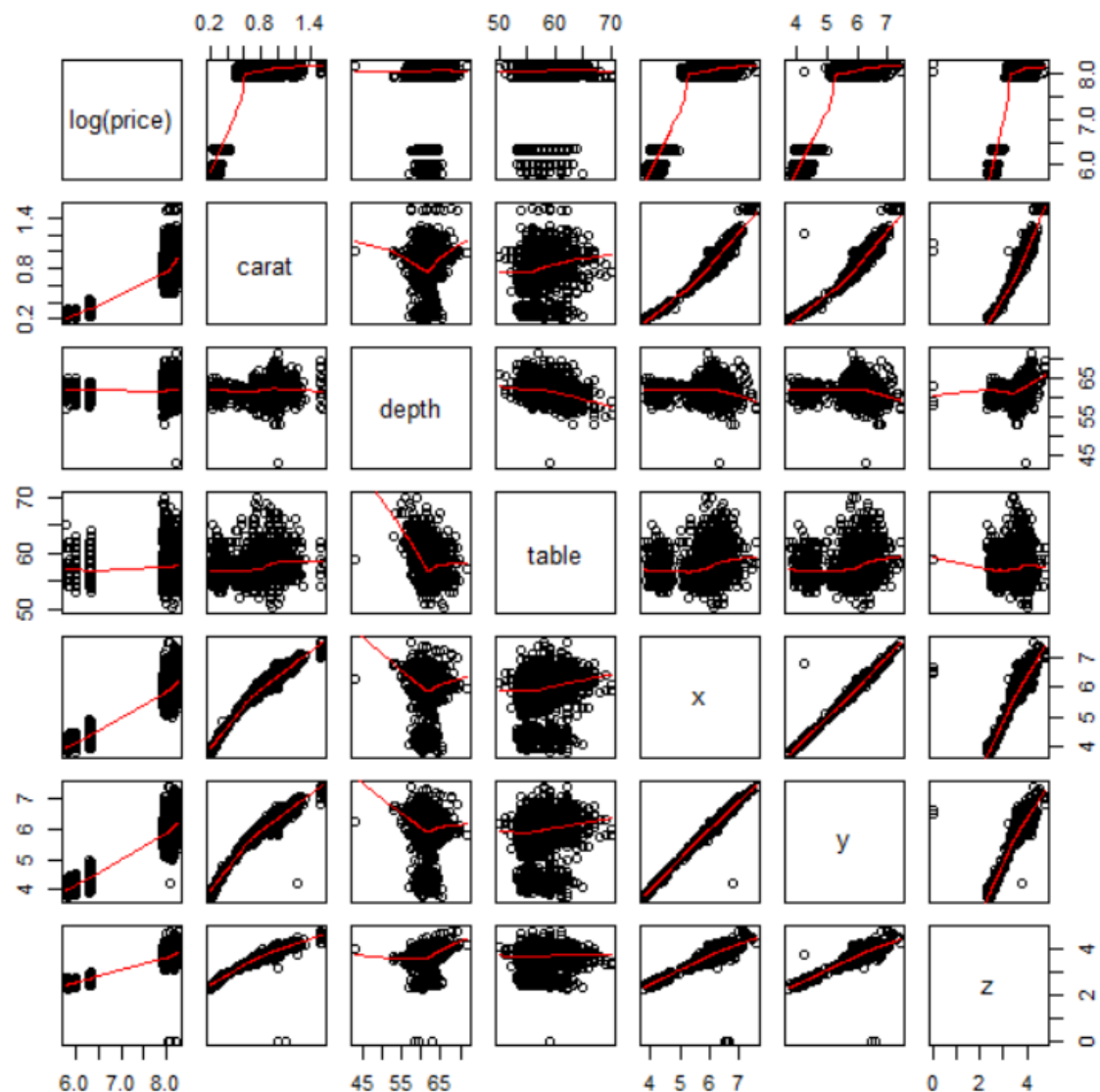
Model Fit

- **Original model:** $R^2 = 0.8557$
- **Log model:** $R^2 = 0.9005$

Summary of differences:

- The **log model fits better** ($R^2 = 90\%$ vs 85.6%).
- In the **original model**, coefficients show *absolute dollar changes*.
- In the **log model**, coefficients show *percentage changes*.
- Both models show significance for all predictors, but the **log model reduces residual skewness** and gives a clearer interpretation.

LOG(PRICE) VS EACH PREDICTORS:



Relationship of predictors with log(price)

- Carat vs log(price):**
 Strong positive and nearly linear relationship. Larger carat strongly increases log(price). This is the dominant predictor.
- x, y, z vs log(price):**
 Also show strong positive, nearly linear relationships, but they overlap heavily with carat. This confirms multicollinearity (x, y, z are just measurements of carat/size).
- Depth vs log(price):**
 No clear linear trend. Shows a slightly curved, flat pattern → weak predictor.
- Table vs log(price):**
 Mostly flat or weakly curved, suggesting very little influence on log(price).

Relationship among predictors

- x, y, z, and carat are almost perfectly correlated → strong multicollinearity problem.
- Depth and Table are uncorrelated with carat, x, y, z → they provide unique information but weak effects.

MULTICOLLINEARITY:

```
> #Multicollinearity
> vif(log_mlr)
      carat      depth      table         x         y         z
37.857823   2.869489   1.256569 129.934616  90.835509  15.046145
```

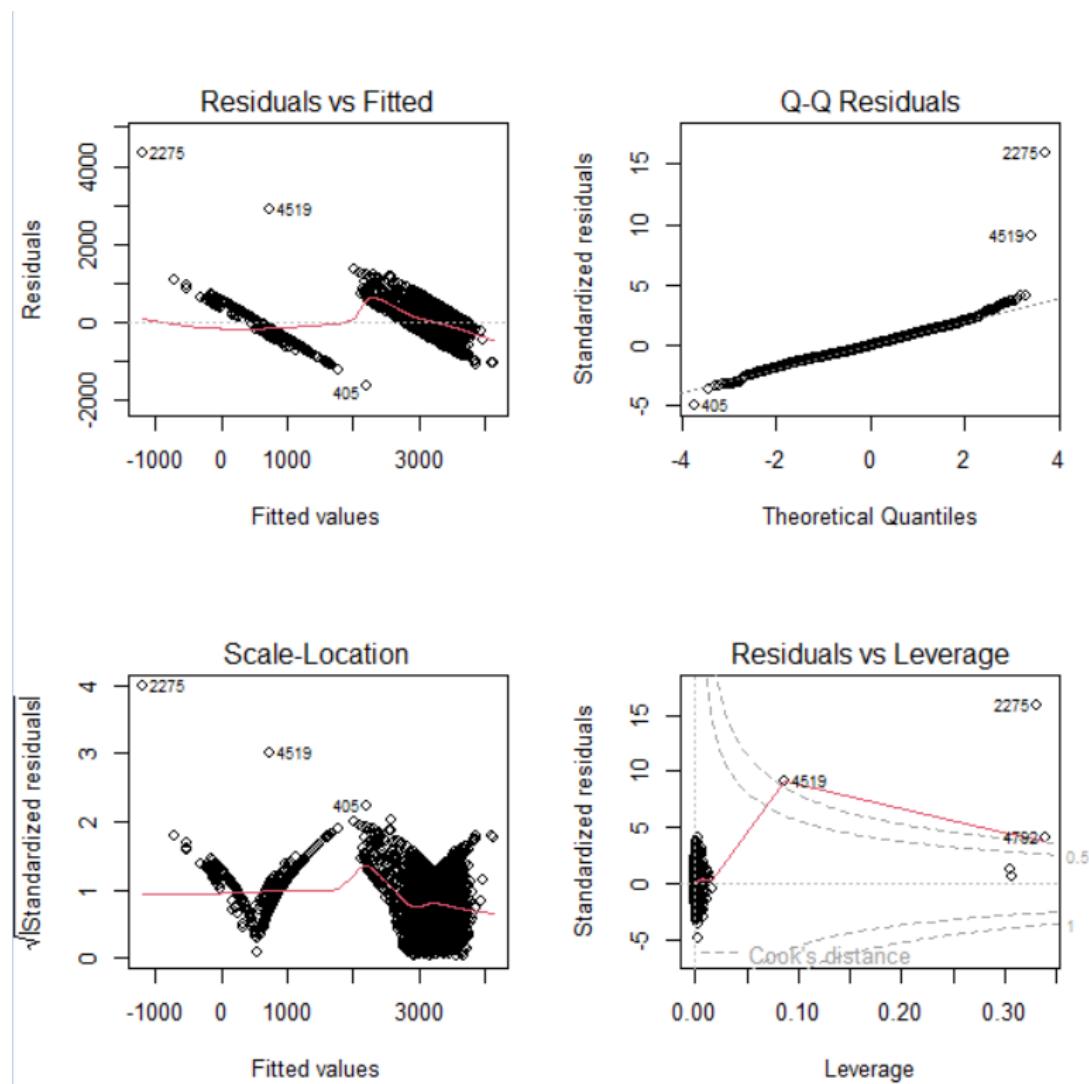
Results

- Carat (37.86), x (129.93), y (90.83), z (15.05) → Extremely high VIFs → severe multicollinearity. These variables are highly correlated with each other (e.g., carat and dimensions).
- Depth (2.87), Table (1.26) → Low VIFs → no multicollinearity issue.

Interpretation

The model suffers from serious multicollinearity among carat, x, y, and z, which means they overlap heavily in the information they provide about diamond size. This inflates standard errors and makes coefficient estimates unstable or even misleading. On the other hand, depth and table are independent and safe.

RESIDUAL PLOTS:



1. Residuals vs Fitted (top-left)

- The red line deviates from horizontal → suggests non-linearity in the relationship (model may not capture all patterns).
- Spread of residuals increases slightly → possible heteroscedasticity (non-constant variance).
- Some extreme points (2275, 4519, 405) indicate outliers.

2. Normal Q-Q (top-right)

- Most points follow the diagonal, but ends deviate → residuals are not perfectly normally distributed.
- Outliers (2275, 4519) deviate strongly from normality.

3. Scale-Location (bottom-left)

- Red line is not flat and variance increases with fitted values → confirms heteroscedasticity.
- Suggests transformation (log on predictors or response — you already did log(price), which helps) or weighted regression.

4. Residuals vs Leverage (bottom-right)

- Points 2275, 4519, and 4792 have high leverage and influence (Cook's distance is large).
- These observations could disproportionately affect the regression model → consider investigating/removing them.

Overall Interpretation:

Your model shows:

- Some non-linearity and heteroscedasticity.
- Residuals are roughly normal, but a few outliers break the assumption.
- High-leverage influential points exist, which may distort coefficients.

Conclusion

The multiple linear regression analysis of the diamonds dataset shows that diamond size is the primary driver of price. Carat, length (x), width (y), and depth (z) all have strong positive and nearly linear relationships with price (especially after applying a log transformation to price, which stabilized variance and improved model fit).

However, because carat, x, y, and z measure similar aspects of diamond size, they exhibit severe multicollinearity, which inflates variance and destabilizes coefficient estimates. Therefore, it is advisable to use carat alone as the main size predictor, or apply dimension reduction techniques such as PCA if including x, y, z.

On the other hand, depth and table do not show strong linear relationships with log(price) and have weaker predictive contributions. While they may add minor information, their effect is limited compared to size-related variables.

Overall, the log-transformed model is more reliable than the raw price model, as it explains a higher proportion of variance, produces more stable coefficients, and allows interpretation in terms of percentage changes in price.

REFERENCE:

1. <https://www.kaggle.com/datasets/roxiroxi/diamond-price-prediction>